

What are Conditional Statements?

Conditional statements are used to make decisions in a program. They allow the program to **execute a block of code only if a certain condition is true**, otherwise execute a different block (or skip execution).

In Python, conditions are written using comparison operators (`==`, `!=`, `<`, `>`, `<=`, `>=`) and logical operators (`and`, `or`, `not`).

1. The if Statement

The simplest conditional is the if statement. It executes a block of code only if the condition is true.

Syntax

if condition:

 statement(s)

Example

```
a = int(input("Enter a number: "))
```

```
if a == 56:
```

```
    print("Right")
```

If the user enters 56, output will be:

Right

2. The if-else Statement

If the condition is true, one block runs; if false, another block runs.

Syntax

if condition:

 statement(s) # executes if condition is true

else:

 statement(s) # executes if condition is false

Example 1

```
a = int(input("Enter a number: "))
```

```
if a == 56:
```

```
    print("Right")
```

```
else:
```

```
    print("Wrong")
```

Example 2: Odd or Even

```
a = int(input("Enter a number: "))
```

```
if a % 2 == 0:
```

```
    print("Even number")
```

```
else:
```

```
    print("Odd number")
```

3. The elif Statement

elif stands for "else if". It is used when there are **multiple conditions** to check.

Syntax

```
if condition1:
```

```
    statement(s)
```

```
elif condition2:
```

```
    statement(s)
```

```
elif condition3:
```

```
    statement(s)
```

```
else:
```

```
    statement(s)
```

Example: Weekdays

```
a = int(input("Enter a number (1-7): "))
```

```
if a == 1:
```

```
    print("Monday")
elif a == 2:
    print("Tuesday")
elif a == 3:
    print("Wednesday")
elif a == 4:
    print("Thursday")
elif a == 5:
    print("Friday")
elif a == 6:
    print("Saturday")
elif a == 7:
    print("Sunday")
else:
    print("Enter weekdays only")
```

4. Nested if Statements

An if statement inside another if statement is called a **nested if**. This is useful when we need to check multiple levels of conditions.

Example: Grace Marks

```
a = int(input("Enter marks: "))
if a < 33:
    print(f"{a} + 4 = {a + 4}")
    a += 4
if a < 33:
    print("Fail")
else:
```

```
    print("Passed by grace marks")
else:
    print("Pass")
```

5. Practice Example: Grading System

```
a = int(input("Enter marks: "))
```

```
if a <= 33:
```

```
    print("Fail")
```

```
elif a > 33 and a <= 50:
```

```
    print("E grade")
```

```
elif a > 50 and a <= 65:
```

```
    print("D grade")
```

```
elif a > 65 and a <= 75:
```

```
    print("C grade")
```

```
elif a > 75 and a <= 85:
```

```
    print("B grade")
```

```
elif a > 85 and a <= 95:
```

```
    print("A grade")
```

```
elif a > 95 and a <= 100:
```

```
    print("A+ grade")
```

```
else:
```

```
    print("Wrong input")
```

6. Example: Youngest Person (Nested if)

```
a = int(input("Enter the age of 1st person: "))
```

```
b = int(input("Enter the age of 2nd person: "))
```

```
c = int(input("Enter the age of 3rd person: "))
```

```
d = int(input("Enter the age of 4th person: "))
```

```
if a < b:
```

```
    if a < c:
```

```
        if a < d:
```

```
            print("1st person is youngest")
```

```
if b < a:
```

```
    if b < c:
```

```
        if b < d:
```

```
            print("2nd person is youngest")
```

```
if c < a:
```

```
    if c < b:
```

```
        if c < d:
```

```
            print("3rd person is youngest")
```

```
if d < a:
```

```
    if d < b:
```

```
        if d < c:
```

```
            print("4th person is youngest")
```

2. Programming Problems (Conditional Statements Only)

1. WAP to check whether a person is eligible for voting or not (age ≥ 18).
2. WAP to check whether a number entered by the user is odd or even.

3. WAP to check whether a number is divisible by 7 or not.
4. WAP to check whether the last digit of a number is divisible by 3 or not.
5. WAP to check whether a year is a leap year or not.
6. WAP to accept percentage from the user and display the grade (A, B, C, D).
7. WAP to accept a number from 1 to 7 and display the name of the day.
8. WAP to accept two numbers and an operator (+, -, *, /) and perform the operation.
9. WAP to check whether a number entered by the user is a three-digit number or not.
10. WAP to find the smallest number out of two numbers.
11. WAP to find the largest number out of two numbers.
12. WAP to check whether a number is divisible by both 2 and 3.
13. WAP to accept the age of 4 people and display the youngest one.
14. WAP to accept the age of 4 people and display the oldest one.
15. WAP to check whether a character is a vowel or not.